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Dynamic Line Rating Assisted Transient Stability Preventive Control with High Penetration of Renewable Energy

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Abstract—With the widespread adoption of renewable energy sources, their efficient utilization is frequently limited by the capacity of transmission lines. This limitation necessitates the implementation of dynamic line rating (DLR). This paper proposes a DLR-assisted transient stability preventive control method in the presence of high penetration of renewable energy. The DLR is computed based on the one-hour-ahead weather forecast and a data-driven deep belief network (DBN) is developed to replace the time-consuming transient stability calculations. The DLR and DBN are integrated to establish line capacity and transient stability constraints in the preventive control model. The proposed method is validated on a modified 39-bus system, demonstrating that DLR can significantly improve the utilization of renewable energy, reduce generation costs and ensure transient stability concurrently.

Index Terms—Dynamic line rating (DLR), deep belief network (DBN), power system transient stability, preventive control, renewable energy integration.

I. INTRODUCTION

WITH the increasing annual load and renewable energy integration, existing transmission lines are often operating close to their capacity limits. Upgrading and retrofitting these transmission lines requires significant investments. The existing widespread use of static line rating (SLR) limits the utilization of the actual transmission line capacity [1]–[3], as it is typically determined based on conservative annual or seasonal weather conditions.

To provide more accurate predictions of line capacity, dynamic line rating (DLR) is introduced. DLR is calculated based on measured or forecasted weather conditions and is typically higher than SLR [1]. The calculation of DLR is defined in IEEE Std. 738 [4] and the CIGRE SC.B2 WG 22.12 [5], with both methods yield identical results, as shown in [6], [7]. According to the analysis in [8] and [9], adopting DLR can enhance network reliability, facilitate greater wind energy integration, and unlock significant economic potential. Several optimal power flow (OPF) methods that take DLR into consideration have been proposed [10], [11]. In [12], a distributionally robust congestion management model is proposed based on a binary rating forecast. This model selectively

employs dynamic ratings on critical lines to maintain thermal overloading risk below a specified threshold. [13] analyzes the potential of DLR and integration into a dispatch optimization process, while effectively managing potential errors in line rating forecasts with centralized and decentralized control approaches. In [7], DLR is incorporated into security-constrained unit commitment, highlighting the improvements in computation time and system security. Additionally, DLR has been considered in transmission expansion planning [14] and cascading failure risk mitigation [15], demonstrating its applications to a variety of optimization, planning, and fault mitigation problems. These applications have proven significant advantages in enhancing reliability, reducing safety risks, and improving economic efficiency. However, those aforementioned work only consider static constraints, which may lead to more severe transient stability failure risks, as discussed in this paper.

In this paper, a DLR-assisted transient stability preventive control method is proposed to address the challenge imposed by the high penetration of renewable energy resources. The key innovation of this work is to demonstrate that the consideration of transient stability in the DLR-constrained OPF will yield more realistic transmission system power transfer results. Existing literature only considers DLR in the static ACOPF problem. However, the DLR only shows the maximum physical power transfer, which does not show the actual power transfer limit under different operational conditions as the transient stability can pose additional restrictions. This paper for the first time demonstrates that the DLR and DBN can be integrated to calculate realistic power transfer of transmission lines while respecting all other operational constraints under the security-constrained OPF framework. The results demonstrate that the proposed method enhances the utilization of renewable energy and reduces generation costs while maintaining the transient stability of the system.

II. PROBLEM STATEMENT

This section introduces the traditional transient stability-constrained OPF (TSC-OPF) for preventive control.

A. Objective Function

The objective function is to minimize the total cost of generation and renewable energy curtailment.

$$\text{Minimize } \sum_{g=1}^{n_g} C_g(P_{g,t}) + \sum_{r=1}^{n_r} C_r(\Delta P_{r,t}) \quad (1)$$

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where $C_g(P_{g,t}) = a_g P_{g,t}^2 + b_g P_{g,t} + c_{g,t}$; $C_r(\Delta P_{r,t}) = c_r \Delta P_{r,t}$; $\Delta P_{r,t} = P_{r,t}^{MPPT} - P_{r,t}$; t denotes current time instant; n_g is the number of synchronous generators; $P_{g,t}$ denotes the active power output of g -th synchronous generator at time t ; $C_g(\cdot)$ is the generation cost function; a_g , b_g , and c_g are cost constants of P_g ; n_r is the number of renewable energy generators; $P_{r,t}$ denotes the active power output of r -th renewable energy generator; $P_{r,t}^{MPPT}$ denotes the maximum active power output of $P_{r,t}$ according to maximum power point tracking (MPPT); $\Delta P_{r,t}$ denotes the curtailed active power of $P_{r,t}$; $C_r(\cdot)$ is the renewable energy curtailment cost function; c_r is cost constant of P_r .

B. Power Balance Constraints

The polar AC power balance constraints are [16]

$$P_{i,t} = \sum_{j=1}^{n_b} G_{ij} V_{i,t} V_{j,t} \cos \theta_{ij,t} + \sum_{j=1}^{n_b} B_{ij} V_{i,t} V_{j,t} \sin \theta_{ij,t} \quad (2)$$

$$Q_{i,t} = \sum_{j=1}^{n_b} G_{ij} V_{i,t} V_{j,t} \sin \theta_{ij,t} - \sum_{j=1}^{n_b} B_{ij} V_{i,t} V_{j,t} \cos \theta_{ij,t} \quad (3)$$

where $i = 1, 2, \dots, n_b$; n_b is the number of buses; $P_{i,t}$ and $Q_{i,t}$ are injected active and reactive power at bus i at time t ; $V_{i,t}$ and $V_{j,t}$ are voltage magnitude at bus i and j ; $\theta_{ij,t}$ is bus voltage angle difference between bus i and j ; G_{ij} and B_{ij} are the real and imaginary parts of the bus admittance Y_{ij} ;

C. SLR-Based Transmission Line Capacity Constraints

The SLR-based transmission line capacity inequality constraints are

$$|P_{l,t}| \leq P_l^{\max}, l = 1, 2, \dots, n_l \quad (4)$$

where n_l is the number of transmission lines; $P_{l,t}$ denotes the active power flow at line l ; $P_l^{\max}$ is the max active power flow limits of the lines.

D. Variable Limits

Variable limits include slack bus voltage angle constraint, bus voltage magnitude constraints, synchronous generator active and reactive power output constraints, and renewable energy generator active and reactive power output constraints, i.e.,

$$\theta_{i,t}^{\text{slack}} \leq \theta_i \leq \theta_i^{\text{slack}}, i \in \zeta^{\text{slack}} \quad (5)$$

$$V_i^{\min} \leq V_{i,t} \leq V_i^{\max}, i = 1, \dots, n_b \quad (6)$$

$$P_g^{\min} \leq P_{g,t} \leq P_g^{\max}, g = 1, \dots, n_g \quad (7)$$

$$Q_g^{\min} \leq Q_{g,t} \leq Q_g^{\max}, g = 1, \dots, n_g \quad (8)$$

$$P_r^{\min} \leq P_{r,t} \leq P_{r,t}^{MPPT}, r = 1, \dots, n_r \quad (9)$$

$$Q_r^{\min} \leq Q_{r,t} \leq Q_{r,t}^{MPPT}, r = 1, \dots, n_r \quad (10)$$

where ζ^{slack} is the slack bus; θ_i and θ_i^{slack} denote the phase angle of slack bus voltage and its rated value. (5) means that the phase angle of the slack bus is constant, and other buses use it as a reference; $V_{i,t}$ denotes the i -th bus voltage magnitude;

$V_i^{\max}$ and $V_i^{\min}$ are upper and lower limits of V_i ; $Q_{g,t}$ denotes the g -th synchronous generator reactive power output; $P_g^{\max}$ and $P_g^{\min}$ are upper and lower limits of P_g ; $Q_g^{\max}$ and $Q_g^{\min}$ are upper and lower limits of Q_g ; $Q_{r,t}$ denotes the r -th renewable energy generator reactive power output; $P_r^{\min}$ and $Q_r^{\min}$ are lower limits of $P_{r,t}$ and $Q_{r,t}$, while $P_{r,t}^{MPPT}$ and $Q_{r,t}^{MPPT}$ are upper limits according to MPPT.

E. Transient Stability Constraint

Transient stability is constrained by differential-algebraic equations (DAEs).

$$\dot{\mathbf{x}}_t = \mathbf{f}(\mathbf{x}_t, \mathbf{y}_t, \mathbf{u}, \mathbf{p}) \quad (11)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}_t, \mathbf{y}_t, \mathbf{u}, \mathbf{p}) \quad (12)$$

where $\mathbf{x}_t$ denotes the state variables, such as the rotor angle and speed of synchronous generators; $\mathbf{y}_t$ denote the algebraic variables, such as voltages and power flows; $\mathbf{u}$ is the input variables; $\mathbf{p}$ denotes the model parameters; $\mathbf{f}(\cdot)$ and $\mathbf{g}(\cdot)$ are nonlinear DAEs, which respectively describe the dynamic and static components of the power system. It should be noted that TSC-OPF is an extension of OPF with the addition of transient stability constraints. The presence of DAEs makes solving the transient stability constraint a time-consuming process and raises scalability concerns, particularly for large-scale systems.

III. PROPOSED TRANSIENT STABILITY PREDICTIVE CONTROL CONSIDERING DLR

This section shows the modeling of DLR and its inclusion into the TSC-OPF for enhancing system line capacity as well as transient stability.

A. Heat Balance Equation-Based DLR Calculation

According to IEEE Std. 738 [4], the temperature of overhead transmission lines depends on: 1) Conductor size and resistance; 2) current flowing through the conductor; 3) ambient weather conditions, such as temperature, wind speed and direction, and solar radiation [7]. The heat balance equation (HBE) is used to describe the thermal energy entering and leaving the conductor, and it can be represented by equation (13), where the left side corresponds to the cooling terms of convection and radiation, while the right side corresponds to the heating terms of solar heat and resistance heat.

$$q_c + q_r = q_s + I^2 R(T_{\text{avg}}) \quad (13)$$

where T_{avg} denotes the average temperature of aluminum strand layers [$^{\circ}\text{C}$]; $R(T_{\text{avg}})$ is the AC resistance of conductor at temperature T_{avg} [Ω/m]; I is the conductor current [A]; q_c , q_r , q_s and $I^2 R(T_{\text{avg}})$ are convection heat loss, radiation heat loss, solar heat gain and resistance heat gain of conductor, respectively [W/m]. The calculation of q_c depends on different wind speeds. When there is zero wind speed, q_c is derived from natural convection, and it is given by

$$q_{cn} = 3.645 \rho_f^{0.5} D_0^{0.75} (T_s - T_a)^{1.25} \quad (14)$$

where ρ_f is the density of air [Kg/m^3]; D_0 is the outside diameter of conductor [m]; T_s denotes the conductor surface temperature [$^{\circ}\text{C}$]; T_a denotes the ambient air temperature [$^{\circ}\text{C}$].

When there is low wind speed or high wind speed, it is determined by the following

$$q_{c1} = K_{\text{angle}} [1.01 + 1.35 N_{\text{Re}}^{0.52}] k_f (T_s - T_a) \quad (15)$$

$$q_{c2} = K_{\text{angle}} 0.754 N_{\text{Re}}^{0.6} k_f (T_s - T_a) \quad (16)$$

where K_{angle} is the Wind direction factor; N_{Re} is the dimensionless Reynolds number; k_f is the thermal conductivity of air at temperature $(T_s + T_a)/2$ [W/(m $\cdot$ °C)]. (15) is accurate for low wind speeds but underestimates forced convection at high wind speeds. Conversely, (16) is correct at high wind speeds but underestimates forced convection at low wind speeds. At any wind speed, IEEE Std. 738 recommends computing q_c using both equations and selecting the larger of the two calculated q_c . q_r can be calculated by

$$q_r = 17.8 D_0 \varepsilon \left[\left(\frac{T_s + 273}{100} \right)^4 - \left(\frac{T_a + 273}{100} \right)^4 \right] \quad (17)$$

where ε is the emissivity. q_s can be calculated by

$$q_s = \alpha Q_{se} \sin(\theta) A' \quad (18)$$

where α denotes the solar absorptivity; Q_{se} denotes the total solar and sky-radiated heat intensity corrected for elevation [W/m 2]; θ is the effective angle of incidence of the sun's rays [deg]; A' is the projected area of conductor [m 2 /linear m]. For each overhead transmission line, the maximum allowable active power for the transmission line can be calculated based on the current weather conditions by

$$P^{\text{DLR}} = \left(\sqrt{\frac{q_c + q_r - q_s}{R(T_{\text{avg}})}} \right)^2 R(T_{\text{avg}}) L = (q_c + q_r - q_s) L \quad (19)$$

where L denotes the length of the overhead transmission line [m]. (19) can be utilized in (4) to establish the DLR-based line capacity constraints for critical lines.

B. DBN-Based Transient Stability Prediction

DBN is composed of a stack of restricted Boltzmann machines (RBM) and is used for transient stability prediction in this paper. The detailed model structure, sample generation and model training for the DBN can be found in [17]. The transient stability index (TSI) is used to represent the stability of the system and is defined as follows

$$\text{TSI} = \frac{360^\circ - \delta^{\max}}{360^\circ + \delta^{\max}} \times 100 \quad (20)$$

$$\delta^{\max} = \max_{t \in T_s} (|\delta_{i,t} - \delta_{j,t}|), \forall i, j \in n_g \quad (21)$$

where T_s denotes the entire duration of TDS; $\delta_{i,t}$ and $\delta_{j,t}$ denote the rotor angles of generator i and j at time t ; $\delta^{\max}$ is the maximum rotor angle difference of any two generators. There is a positive correlation between transient stability and TSI. The proposed method is capable of handling multiple contingencies by controlling only the most severe contingency. The DBN-based transient stability prediction can be expressed by

$$\text{TSI}_P = \text{DBN}(\mathcal{I}, \Theta) \quad (22)$$

where $\mathcal{I} = [P_r, P_g, P_d, Q_d]^T$ denotes the input matrix of DBN; P_d and Q_d denote the active and reactive demands; Θ denotes all parameters of DBN, including weights and biases.

To ensure the reliability of the control, the boundary between transient stable and unstable is defined as $\delta^{\max} = 180^\circ$, which can be expressed as $\text{TSI}_B = 33.3$. Correspondingly, the transient stability constraint (11) and (12) can be replaced with DBN as follows

$$\text{DBN}(\mathcal{I}, \Theta) > \text{TSI}_B \quad (23)$$

C. Transient Stability Preventive Control Considering DLR

Taking into account the DLR-based transmission line capacity constraints and the DBN-based transient stability constraint, the proposed data-driven transient stability preventive control framework can be expressed as follows

$$\text{Minimize (1)} \quad (24)$$

$$\text{s.t. (2) - (10)} \quad (25)$$

$$|P_{c,t}| \leq P_c^{\text{DLR}}, c = 1, \dots, n_c \quad (26)$$

$$\text{DBN}(\mathcal{I}, \Theta) > \text{TSI}_B \quad (27)$$

where n_c is the number of critical lines that consider DLR. The proposed method involves a non-linear, non-convex optimization problem that can be solved using an interior point method (IPM). The overall workflow of this paper is illustrated in Algorithm 1.

Algorithm 1: Overall workflow of the proposed method

Input: Parameters of systems; one-hour-ahead weather forecast

Output: Preventive control strategy

- 1 Calculate the DLR for each critical line based on the one-hour-ahead weather forecast
 - 2 Generate random discrete samples of P_r , P_g , P_d and Q_d based on their range limits
 - 3 Train DBN and test performance
 - 4 Build the DBN-based TSC-OPF considering DLR
 - 5 Solve the proposed optimization problem by using IPM and get the preventive control strategy.
-

IV. NUMERICAL RESULTS

A. Test System

The proposed method is validated using the modified IEEE 39-bus system, which includes 6 synchronous generators and 4 wind generators. The system is divided into three areas, with area 1 being characterized as a wind power-rich area, consisting entirely of wind generators. Area 1 supplies renewable energy to areas 2 and 3 via lines 16-17 and 15-16, respectively, making them critical lines. The modified IEEE 39-bus system is illustrated in Fig. 1.

B. DLR Calculation

Based on hourly measurements of weather conditions (ambient temperature and wind speed) for the transmission lines, the DLRs for two critical lines are calculated and compared to the SLR in Fig. 2. Additionally, the daily load and wind power curves are also presented in Fig. 2.

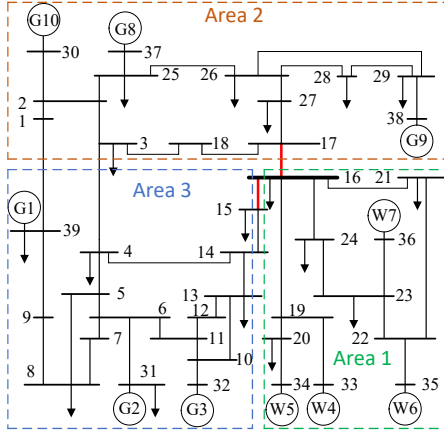


Fig. 1. Modified IEEE 39-bus system (G: synchronous generators; W: wind generators).

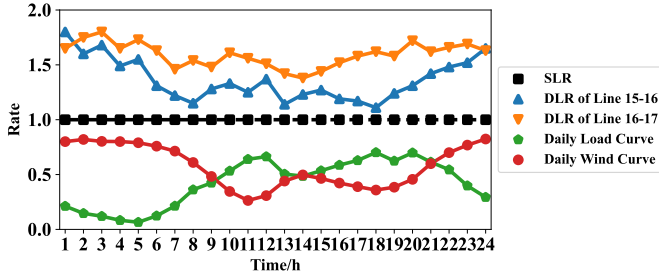


Fig. 2. SLR, DLR, and daily load and wind curves.

C. DBN for Transient Stability Prediction

The system model is constructed in PowerWorld. The wind generators are modeled via DFIG. The variations in P_r , P_g , and P_d range from 0% to 150%, 50% to 120%, and 60% to 100% of their rated values, respectively. The power factor of demands remains constant. In the scenario, where wind power output is at its maximum and synchronous generator output is at its minimum, the proportion of wind power to total generation can reach a maximum of 64.9%. The fault type is set to three-phase short-circuit faults for all credible contingencies. Then, 10000 samples and corresponding TSI are generated by using TDS, 8000 samples of which are training set and 2000 samples being test set. The significant accuracy advantage of DBN in transient stability prediction, compared to other machine learning algorithms, has been extensively compared in [17], and therefore, further comparisons are not conducted in this study. By training the DBN, it can perform fast and accurate predictions of transient stability, serving as a replacement for TDS. The training time and prediction time for DBN are approximately 200s and 0.1s, respectively.

D. Preventive Control Considering DLR

Using the DLR for the two critical lines, the daily load and wind curves, and the DBN-based transient stability constraint, an hourly preventive control is executed for one day. Two different scenarios of TSC-OPF are compared: one considers SLR, and the other considers DLR. The active power outputs

of four wind generators are shown in Fig. 3. From Figure 3, it

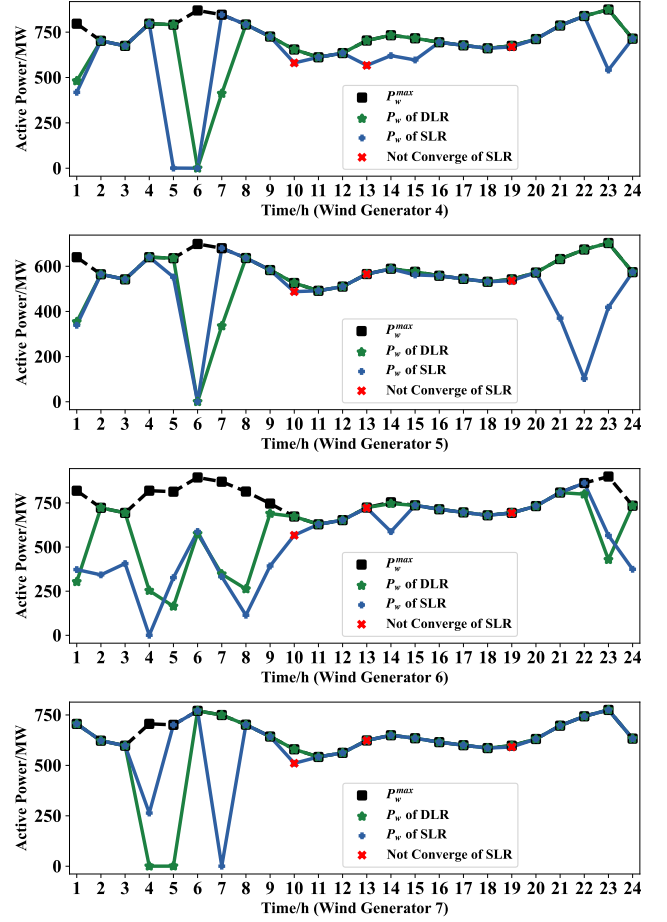


Fig. 3. Comparison of the active power output of wind generators in terms of curtailment.

is evident that the curtailed power of SLR-based TSC-OPF is considerably higher than that of DLR-based TSC-OPF. This observation underscores the significant enhancement in the utilization of renewable energy achieved through the adoption of DLR. Additionally, it should be noted that SLR-based TSC-OPF experiences convergence issues at the 10th, 13th, and 19th hours, while the adoption of DLR consistently ensures effective convergence. The calculation time for preventive control is about 10-30s. The active power of the critical lines is compared in Fig. 4. Fig. 4 indicates that, due to SLR, the active power of the critical lines is restricted; however, DLR-based TSC-OPF allows for the transmission of more renewable energy. This is also the reason for the non-convergence of SLR-based TSC-OPF.

Subsequently, the generation costs of both methods are compared with TSC-OPF without line capacity constraints, as depicted in Fig. 5. Fig. 5 indicates that the cost associated with DLR-based TSC-OPF is lower than that of SLR-based TSC-OPF, only slightly higher than the TSC-OPF that does not consider line capacity constraints. This implies that considering DLR can offer economic advantages.

Finally, in addition to the two methods mentioned above, an OPF considering only DLR without considering transient

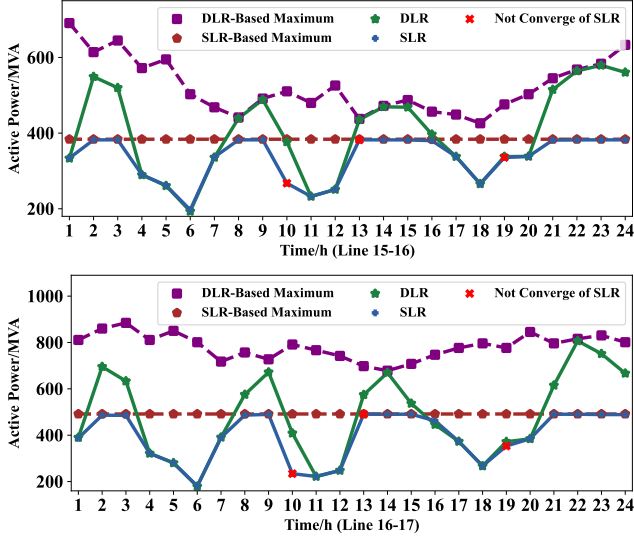


Fig. 4. Comparison of active power of transmission lines using DLR and SLR approaches.

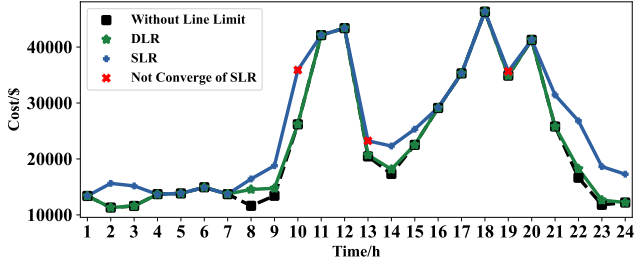


Fig. 5. Comparison of operational cost for each method.

stability constraint is included to highlight the significance of TSC-OPF. The comparison results are shown in Fig. 6. From Fig. 6, it can be observed that when transient stability

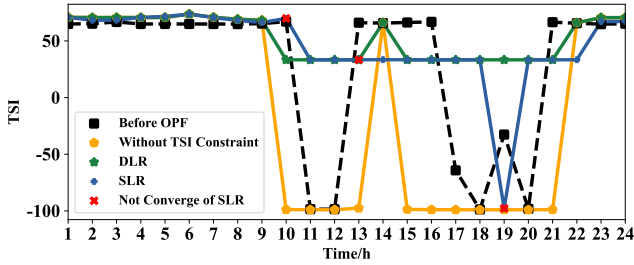


Fig. 6. Comparison of TSI for different approaches.

constraint is not included in the OPF, there is a higher probability of transient instability occurring, as indicated by the yellow line. Therefore, this paper utilizes TSC-OPF as a replacement for traditional OPF to mitigate such risks. After incorporating transient stability constraint, DLR-based TSC-OPF can sustain transient stability, whereas SLR-based TSC-OPF can only achieve transient stability under convergent conditions. This also indicates that DLR can enhance the convergence of the TSC-OPF.

V. CONCLUSION

This paper proposes a DLR-assisted transient stability preventive control method to address the challenges associated with high penetration of renewable energy resources. DLR is employed to establish transmission line capacity constraints for critical lines, while DBN is utilized as a substitute for TDS calculations in transient stability prediction. This method accelerates the solution process. The proposed method is validated on a modified 39-bus system, demonstrating that DLR can significantly improve the utilization of renewable energy, reduce generation costs, and ensure transient stability concurrently. Our future work will involve testing the developed method using actual Eversource Energy system model and real-world data. Furthermore, the uncertainty associated with renewable energy will be considered.

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